Entity Relationship (ER) Diagram of Note Sharing Portal

Introduction to ER Diagram

An **Entity Relationship (ER) Diagram** is a high-level conceptual data model used to visually represent the structure of a database system. It defines:

- Entities (real-world objects)
- Attributes (properties of entities)
- Relationships (connections between entities)
- Cardinality (1:1, 1:M, M:N)

In this Note Sharing Portal system, the ER diagram models how users upload, download, and manage notes within the system.

Entities and Their Attributes

Based on the given ER diagram, there are three main entities:

1. User Entity

The **User** entity represents students who use the system to upload and download notes.

Attributes:

- **User_ID** (Primary Key) – Unique identifier of the user
- UserName – Name of the user
- Email – User's email address
- Password – User's login password

The primary key ensures that each user is uniquely identified in the system.

2. Notes Entity

The **Notes** entity represents the study materials uploaded by users.

Attributes:

- **Note_ID** (Primary Key) – Unique identifier of the note
- Title – Title of the note
- Upload_date – Date when the note was uploaded
- User_ID (Foreign Key) – References the user who uploaded the note

The foreign key (User_ID) establishes a relationship between Notes and User.

3. Admin Entity

The **Admin** entity represents the administrator who manages the notes in the system.

Attributes:

- **Admin_ID** (Primary Key) – Unique identifier of admin
- Name – Admin's name
- Email – Admin's email
- Password – Admin's login password

Admin controls and monitors system activities.

Relationships and Cardinality

◇ Uploads Relationship (User — Notes)

- One User can upload many Notes.
- Each Note is uploaded by only one User.
- Cardinality: **1 : M**

This relationship ensures that every note has an owner.

◇ Download Relationship (User — Notes)

- Many Users can download many Notes.
- Cardinality: **M : N**

This is a many-to-many relationship because:

- A user can download multiple notes.
- A note can be downloaded by multiple users.

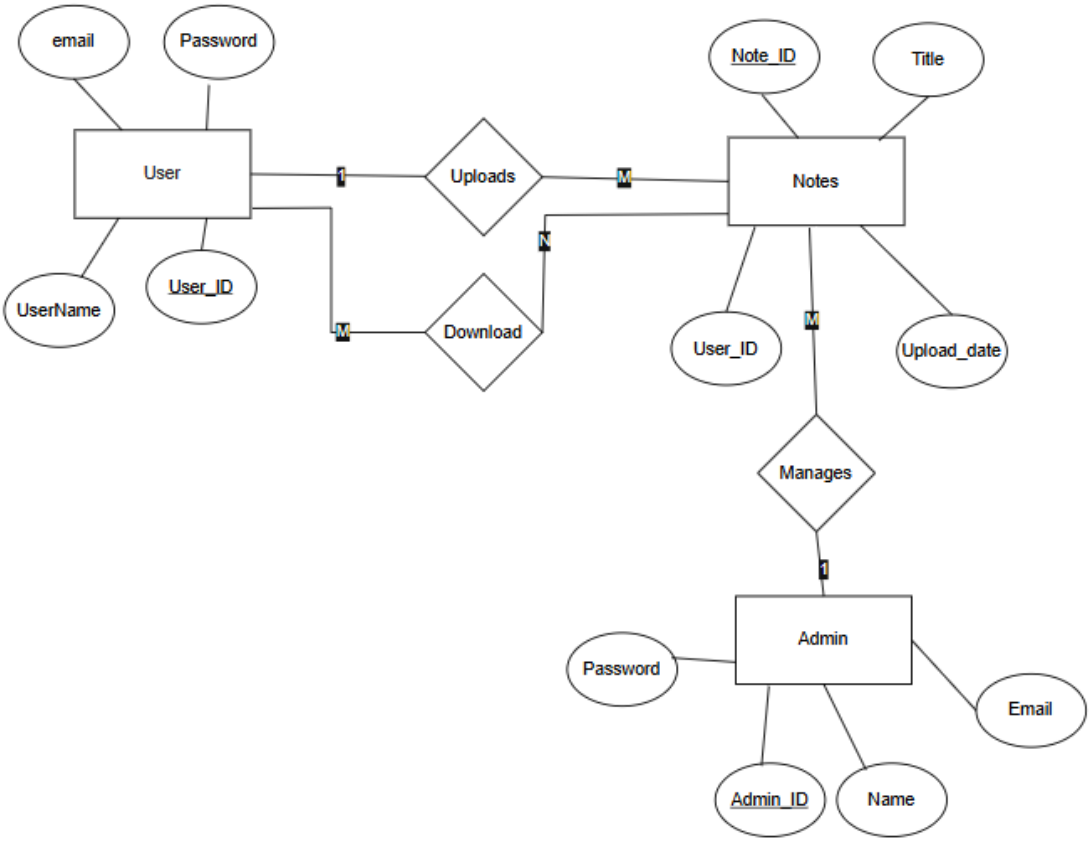
This relationship can be implemented using a separate associative table (Download).

◇ Manages Relationship (Admin — Notes)

- One Admin can manage many Notes.
- Each Note is managed by one Admin.
- Cardinality: **1 : M**

This relationship allows admin to:

- Monitor uploaded notes
- Delete inappropriate notes
- Maintain system integrity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